

Barriers to a European Banking Union

Damien Capelle Adriano Fernandes J.J. Krüger Peter McAdam

IMF

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Disclaimer: The views expressed in the paper are solely those of the authors and do not necessarily represent the views of the IMF or the ECB, its Executive Board, or its Management

Motivation

Policy discourse (Enria 2021; Lagarde 2024; Draghi 2024):

- ▶ European banking markets remain fragmented along national lines
- ▶ *De jure* fragmentation in regulatory frameworks across countries
- ▶ Incomplete integration restrains cross-border credit, constraining growth

Our three questions:

- 1 How large are *de facto* barriers to cross-border banking?**
- 2 Does fragmentation in regulatory frameworks play a role?**
- 3 What are the macroeconomic implications?**



Eurozone economy

“ Brexit and Covid harden the case for a proper EU financial market

Efforts to create banking and capital markets unions have languished for too long



Martin Sandbu



European companies

“ Banking union is essential if the EU is to ride out future crises

The sector must be helped to play its part in delivering Europe's green and digital transition

Paschal Donohoe



Eurozone economy

“ The FT View. A chance to press on with EU banking union

Agreement on bolstering crisis-fighting tools shows that solidarity holds

The editorial board



Europe Express

A road map to nowhere

Eurozone struggles to salvage hopes for stalled Banking Union

PREMIUM



Eurozone economy

Eurozone chief seizes on virus and Brexit to revive banking union

Paschal Donohoe to renew efforts for deposit insurance scheme across single currency area



Eurozone economy

EU recovery fund deal revives hopes for eurozone banking union

Central bankers welcome joint debt issuance agreement as a potential breakthrough



EU financial regulation

EU set to remove barriers to banks' cross-border capital flows

Draft report by European Commission aims to boost performance of lenders in bloc compared with US rivals



EU financial regulation

EU resists banks' plea for major overhaul of capital rules

European Banking Authority sets out "targeted" plan to simplify overlapping requirements for lenders



EU business regulation

“ European banking needs a Big Bang

Creating new rules for cross-border lenders would speed economic recovery



Axel Weber

Paper Overview

1. **Stylized facts** documenting cross-border fragmentation of lending to NFC
 - ▶ Using Euro Area credit registry + group structure (ECB)
2. **Micro-econometric framework to estimate cross-border barriers**
 - ▶ Identification: within-firm/bank comparisons in lending network (Kline 2024)
 - ▶ Key finding: barriers are large and primarily extensive-margin
3. **Novel dataset on banking policy distances** across EA countries
 - ▶ Validation: significant correlations even after accounting for non-policy forces
4. A **GE spatial banking model to quantify GDP gains** from lifting barriers
 - ▶ Capture granularity of within-country firm and bank heterogeneity
 - ▶ Keeping branch network fixed, cross-border barriers $\downarrow$ by 10% $\Rightarrow$ EA GDP + 1.6%

Literature

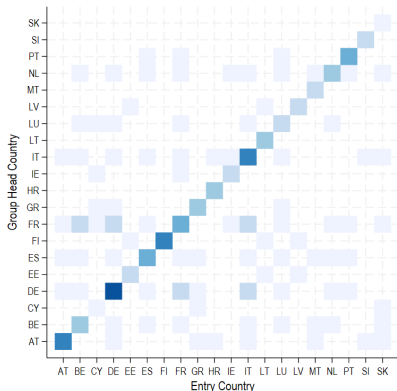
1. **Cross-border financial intermediation:** Buch (2005), Cetorelli and Golderg (2012), Niepmann (2016, 2023), Correa et al (2022), Beck et al (2024), Altavilla et al (2024), Kosekova et al (2024), Degryse et al (2025)
2. **Spatial models of financial intermediation:** Oberfield et al (2025), D'Amico (2025), Aguirregabiria et al (2025), Herreno (2025), Ji et al (2023), Morelli et al (2024)
3. **Wedges in international finance:** Gourinchas and Jeanne (2013), Garleanu et al (2020), Ohanian et al (2018, 2021), Pellegrino et al (2025), Capelle and Pellegrino (2025)
4. **Financial flows and allocative efficiency:** Gilchrist et al (2013), Midrigan and Xu (2014), Moll (2014), Gopinath et al (2017), Saffie et al (2020), Whited and Zhao (2021), Bau and Matray (2023), Bertaut et al (2024), Camelo (2024), Garcia-Santana et al (2025)

Stylized Facts

Facts on Cross-Border Banking: Data

- ▶ **AnaCredit:** near-universe of credit instruments by EA banks, 2019:Q4–2025:Q3
 - ▶ Relationship-level data on loan volumes, interest rates, and characteristics
 - ▶ Complemented by bank balance sheets from iBSI
- ▶ **RIAD:** near-universe of EA banking group structures
 - ▶ Custom algorithm assigns each bank to its ultimate deposit-taking parent
- ▶ **Two definitions of cross-border:**
 - ▶ *Geographic:* location of lender and borrower — where lending technologies and operations are
 - ▶ *Nationality:* ownership of banking group — captures where banking groups are headquartered versus where they branch

European Corporate Lending Markets Are Fragmented



Number of branches by group head $\times$ entry country

1. Cross-border bank loans to NFCs $\approx$ 1% of loans, 5.6% of loan volumes
 - Most **home-biased** asset class, uneven across countries [Plot](#)
2. Almost **no cross-border entry** (branches or subs)
3. In average country pair, **one in every four regulatory variables differ** [Plot](#)

[Dataset](#)

A Micro-Econometric Framework to Estimate Cross-border Wedges

Micro-Econometric Framework (Bank-Firm Level)

Idea: non-parametrically recover cross-border frictions for each country-pair

1. **Relationship formation** (do bank and firm match?):

$$Y_{bft}^L = \exp \left(\alpha_b^L + \alpha_f^L + \alpha_t^L + \theta^L(i_f, i_b) \right) \cdot \varepsilon_{bft}^L \quad [\text{PPML}]$$

2. **Interest rates** (pricing conditional on relationship):

$$\log R_{bft} = \alpha_b^R + \alpha_f^R + \alpha_t^R + \theta^R(i_f, i_b) + \varepsilon_{bft}^R \quad [\text{OLS}]$$

- ▶ Motivated by structural model
- ▶ **Computation:** choice-based sampling + IPW (Manski 1977, Wooldridge 1999)
- ▶ **Identification:** within-firm & within-bank $\rightarrow$ purge composition (Kline 2024)

Result 1: Enormous Barriers to Relationship Formation

- ▶ Summarize average effect with single cross-border (geographic) dummy
- ▶ Cross-border relationships are over 99% less likely than domestic
- ▶ Micro wedges are *larger* than aggregate data would suggest ($\approx 95\%$)

Relationship Formation		
	(1)	(2)
Cross-Border	-8.057*** (0.27)	-12.147*** (0.32)
N	85.5 M	85.5 M
Banks	2,5K	2,5K
Firms	462K	462K
Bank Country FE	Y	N
Firm Country FE	Y	N
Bank FE	N	Y
Firm FE	N	Y
Time FE	Y	Y
Estimator	PPML	

Result 2: Modest Frictions Conditional on Relationships

- ▶ Once relationships form, cross-border loans are ≈ 28 bps cheaper
- ▶ No discount (actually, premium) if exclude factoring
- ▶ Once relationships form, cross-border loans are $\approx 14\%$ smaller [Table](#)

Barriers operate mostly on extensive margin (whether relationships form)

Interest Rates		
	(1)	(2)
Cross-Border	-0.00867*** (0.00077)	-0.00284*** (0.00059)
N	8.9M	8.9M
Banks	2.4K	2.4K
Firms	476K	476K
Dep. Mean	0.033	0.033
Bank Country FE	Y	N
Firm Country FE	Y	N
Bank FE	N	Y
Firm FE	N	Y
Time FE	Y	Y
Estimator	OLS	

Result 3: Large Frictions to Foreign Branches/Subsidiaries

Decisions to open ***branches/subsidiaries*** by banking groups

$$Y_{g(b)it}^E = \alpha_g^E + \alpha_i^E + \alpha_t^E + \theta^E(o_g, i_b) + \varepsilon_{ig}^E \quad [\text{LPM}]$$

$\theta^E(o_g, i_b) = 1$ if nationality of group $\neq$ location of bank originating the loan

Branches/Subsidiaries		
	(1)	(2)
Cross-Border	-0.99*** (0.005)	-0.99*** (0.005)
N	748K	748K
Time FE	Y	Y
Entry Country FE	Y	Y
Group Country FE	Y	N
Group FE	N	Y
Estimator	OLS	

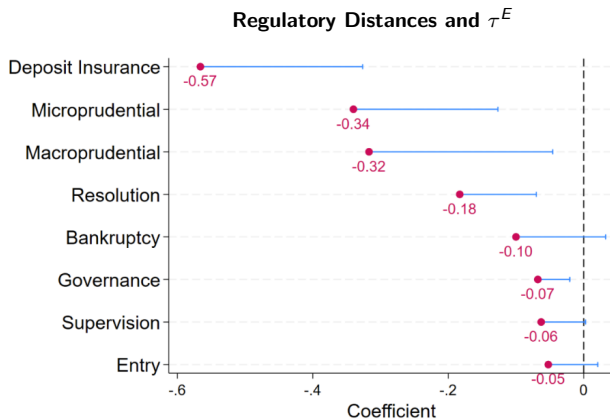
Validation: Policy and Non-Policy Distances

Regulatory Distances Correlate with Branch/Subsidiary Wedges

- ▶ Regulatory fragmentation
⇒ compliance costs
- ▶ Validation: correlate bilateral wedges with regulatory distances

$$\tau_{ij}^L = \beta \cdot \text{Policy}_{ij} + \gamma \cdot \text{Non-Policy}_{ij} + v_{ij}$$

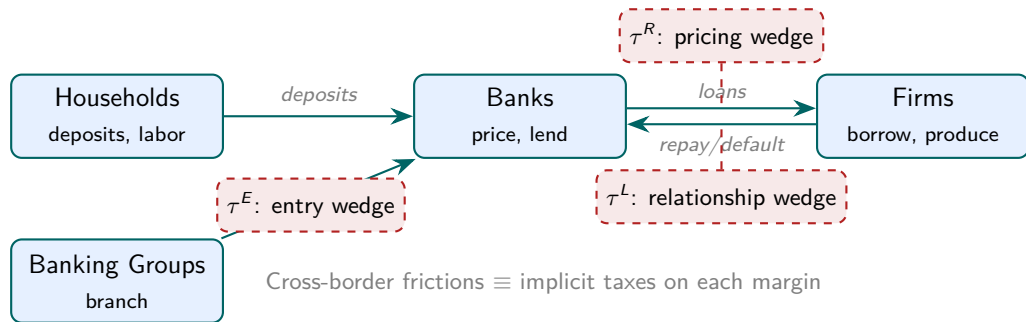
- ▶ Non-Policy_{ij}: geographic, language, cultural proximity (Pellegrino et al., 2025), Orig/Dest FEs



Policy fragmentation significantly related to cross-border entry wedges

A Quantitative General Equilibrium Spatial Model of Banks and Firms

A Spatial GE Model of Cross-Border Banking



- ▶ Multi-country, heterog. firms & banks
- ▶ Firms choose banks
- ▶ Banks set markups on loans, markdowns on deposits
- ▶ Banking groups decide where to branch
- ▶ Microfound the regressions
- ▶ GE: endogenous labor supply and deposits, imperfectly open capital account with RoW, labor mobility

Households

Preferences

$$\max_{c_h, n_h, b \in \mathcal{B}} \ln \left(c_h - \psi_2 \frac{n_h^{1+\psi_3}}{1+\psi_3} \right) \quad \text{s.t.} \quad c_h = (1 + r_b^D) d_h + w n_h + T$$

$$b^* = \operatorname{argmax}_{b \in \mathcal{B}} \left\{ \frac{r_b^D}{\omega_{bh}^D} \right\}$$

Labor supply

$$N^s(w) = \left(\frac{w}{\psi_2} \right)^{1/\psi_3}$$

Deposit choice — logit over banks:

$$p_{ib}^D = \frac{\mathcal{A}_{ib}^D \exp\{\sigma^D r_b^D\}}{\sum_{b' \in \mathcal{B}_i} \mathcal{A}_{ib'}^D \exp\{\sigma^D r_{b'}^D\}}$$

Firms

Technology

$$y_f = z_f k_f^\kappa n_f^\lambda, \quad \kappa + \lambda < 1$$

Financing constraint

$$\sum_b \ell_{bf} = q k_f$$

Profit

$$\pi_f(z_f, G_f) = \max_{n_f, k_f, \ell_{bf}} \left\{ y_f - w n_f - \sum_b R_{bf}^L \ell_{bf} + q(1 - \delta) k_f + G_f \alpha \sum_b R_{fb}^L \ell_{bf} \right\}$$

where G_f is probability of default and α is loss given default

Match quality: specialization (Paravisini, 2023) and monitoring costs (Townsend, 1979)

$$\min_{b \in B} \left\{ \frac{u_{bf}}{\omega_{bfp}^L} \right\} \quad \text{with} \quad u_f = R_{bf}^L (1 - \alpha G_f) + \delta - 1$$

Firms (cont'd)

Loan demand — Fréchet aggregator:

$$p_{bf} = \frac{\mathcal{A}_{bf}^L (u_{bf})^{-\sigma^L}}{\sum_{b'} \mathcal{A}_{b'f}^L (u_{b'f})^{-\sigma^L}} \quad \text{with} \quad \mathcal{A}_{bf}^L = (1 - \tau^L(i_f, i_b)) \tilde{\mathcal{A}}_{bf}^L$$

Composite user cost:

$$u_f^L = \left(\sum_b \mathcal{A}_{bf}^L (u_{bf}^L)^{-\sigma^L} \right)^{-1/\sigma^L}$$

Optimal capital

$$k_f = z_f^{\frac{1}{1-\kappa-\lambda}} \left(\frac{\kappa}{u_f} \right)^{\frac{1-\lambda}{1-\kappa-\lambda}} \left(\frac{\lambda}{w} \right)^{\frac{\lambda}{1-\kappa-\lambda}}$$

Banks

Expected profits (net of wholesale rate R_W):

$$\pi_b^B = [R_b^L - R_W] L_b - [R_b^D - R_W] D_b$$

Loan markup (FOC on R_b^L)

$$R_b^L = \frac{R_W}{1 + \varepsilon_{L/R_b^L}^{-1}} > R_W$$

$\varepsilon_{L/R_b^L} < 0$; loan demand elasticity

Deposit markdown (FOC on R_b^D)

$$R_b^D = \frac{R_W}{1 + \varepsilon_{D/R_b^D}^{-1}} < R_W$$

$\varepsilon_{D/R_b^D} > 0$; deposit supply elasticity

Firm-specific rate:

$$R_{bf}^L = R_b^L \frac{\mathcal{A}_{bf}^R}{1 - \alpha G_f} \quad \text{with} \quad \mathcal{A}_{bf}^R = (1 + \tau^R(i_f, i_b)) \tilde{\mathcal{A}}_{bf}^R$$

Bank Entry: Branches and Subsidiaries

Idea: Banking group g (headquartered in country o) decides whether to open a branch or subsidiary in destination i .

Simplifying assumption: independence across destinations

► in general not true but **cannibalization** turns out quantitatively weak (more later)

Entry probability — LPM:

$$P(\mathbf{1}_{gi} = 1) = F_{\varepsilon}(\pi_{gi}^B > \kappa_{gi}^E) \quad \text{with} \quad \kappa_{gi}^E = \tau^E(i, o_g) + \nu_i + \mu_g + \varepsilon_{gi}^E.$$

Quantitative Exercise

Counterfactual Description and Sources of Gains

Estimation: Calibration + Model Inversion + match micro-econometric wedges

- ▶ State space: all banks + 100 firm bins per country

Counterfactual: Lift cross-border wedges

- ▶ 10% decline in relationship wedges (τ^L) holding network of banks constant Rationale

Sources of gains:

1. Increase in inputs: decrease in $u_f \Rightarrow$ demand for K and $L \uparrow \Rightarrow Y^* \uparrow$
2. Re-allocation through changes in distribution of user costs u_f :

$$Y \approx Y^* - \frac{\rho K}{\kappa u^*} \text{Var}_{w^*}(u_f)$$

Integration Leads to Aggregate GDP Gains

Credit

	Y	K	N	TFP	$\text{Var}(Y/N)$
$\Delta \log (*100)$	+1.61	+2.36	+1.61	+0.06	-2.71
Contribution (%)	100	32.2	64.0	3.8	—

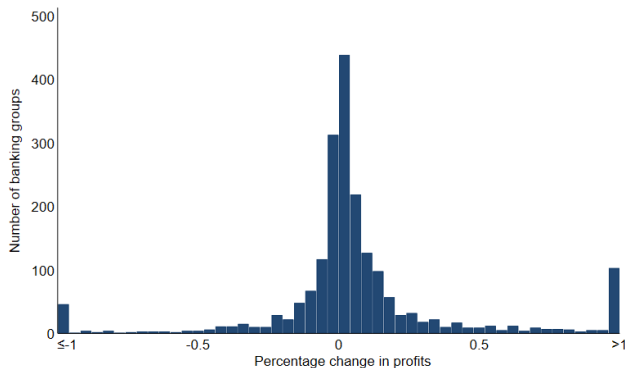
Output gain of **1.6%** driven primarily by capital deepening and increase in labor supply, not by improvements in allocative efficiency

Het: Y

Het: K/L

Misallocation

Endogenous Branching: Very Limited Within-Group Spillovers



- ▶ For each group, add a branch in all countries where currently absent
- ▶ $\mathcal{A}_{g(b)\mathcal{F}_i}^X = \sqrt{\mathcal{A}_{g(b)\mathcal{F}_g}^X \bar{\mathcal{A}}_{b\mathcal{F}_i}^X}$
- ▶ Compute change in profits of incumbents branches
- ▶ **Very limited change in profits** $\Rightarrow$ approx independent decisions
- ▶ **Intuition: home bias + atomistic**
- ▶ Validate independence assumption in empirics and model

Conclusion

Conclusion and Takeaways

1 **Barriers are large and primarily extensive-margin** (relationship, entry)

Micro data reveals more than aggregates

2 **Regulatory fragmentation matters**

Significant correlations even after accounting for non-policy forces

3 **Sizable and heterogeneous gains** across countries.

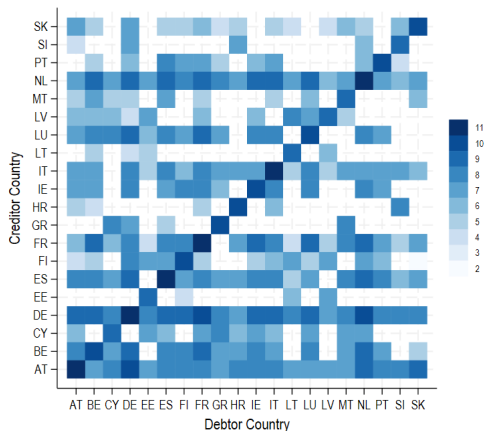
Primarily driven by increases in capital and labor rather than re-allocation

Next steps:

- ▶ Empirics: within-country distance (✓), border design (✓), role of trade receivables, micro vs macro wedges
- ▶ Model: counterfactuals with endogenous branching, frictional labor mobility

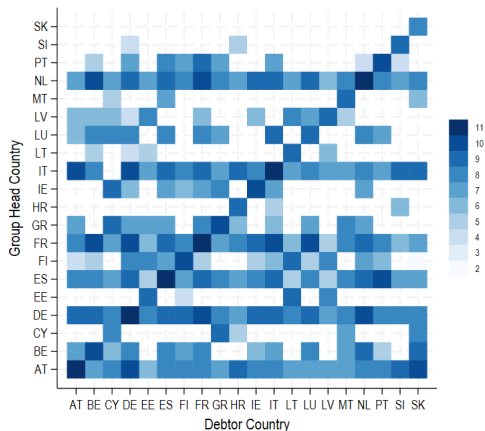
Appendix

Fact 1: Cross-Border (*Geographic*) Lending to NFCs Is Limited

[Nationality](#)[Back](#)

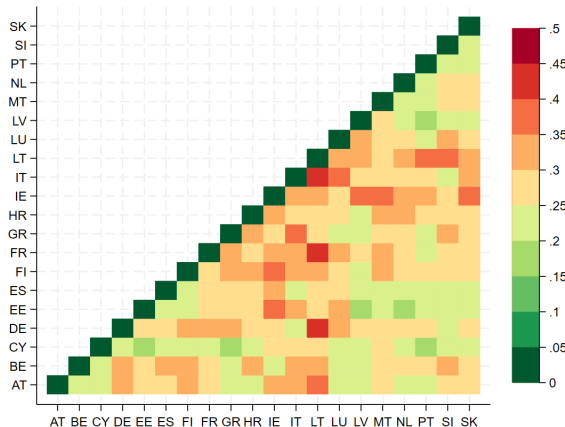
- ▶ Cross-border bank lending to NFCs
≈ **1%** of loans since 2019
≈ **5.6%** of total loan amount
- ▶ Smallest foreign share across intermediaries and assets
- ▶ Huge **home bias**, highly **uneven**

Fact 2b: Cross-Border (*Nationality*) Lending to NFCs Is Limited

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- ▶ Lending volumes mirror sparse entry patterns
- ▶ Cross-border (nationality) bank lending to NFCs
≈ **11%** of loans since 2019
≈ **14.6%** of total loan amount

Fact 3: Regulatory Fragmentation, Despite Harmonization Efforts [Back](#)



Average policy distance by country pair

- ▶ **Gower distance** between each country pair
- ▶ On average, $\approx$ **25%** of policies differ across any two countries [Breakdown](#)

Policy Distances Dataset: Sources

- ▶ Collect 500+ variables on banking regulations
- ▶ **Microprudential & Macprudential:** EBA Options & Discretions + Liquidity Waivers; ESRB capital buffers, borrower-based measures, and reciprocity matrix
- ▶ **Deposit Insurance:** EBA Deposit Guarantee Schemes; IADI annual survey (2014–2024)
- ▶ **Supervision, Resolution, Entry & Governance:** World Bank Bank Regulation & Supervision Survey
- ▶ **Corporate bankruptcy:** OECD Insolvency Indicators (2010, 2016, 2022)

- ▶ Once relationships form, cross-border loans are $\approx 12.5\%$ smaller
- ▶ Effect vanishes when excluding trade receivables

Loan Quantities		
	(1)	(2)
Cross-Border	-0.600*** (0.04937)	-0.1257* (0.07296)
N	8.9M	8.9M
Banks	2.4K	2.4K
Firms	476K	476K
Dep. Mean	0.437	0.438
Bank Country FE	Y	N
Firm Country FE	Y	N
Bank FE	N	Y
Firm FE	N	Y
Time FE	Y	Y
Estimator	OLS	

Market Clearing

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$$\text{Loans:} \quad L_b = \int_f p_{bf} \ell_f df$$

$$\text{Deposits:} \quad D_b = \int_h p_{bh}^D d_h dh$$

$$\text{Labor:} \quad \int_f n_f df = N^s(w) \text{ (closed, perfect mobility within EA)}$$

$$\text{Capital:} \quad \int_f k_f df = K^s(R_W) \text{ (imperfect openness with RoW)}$$

By Walras' law, goods market clears

Counterfactual Effects of Removing Cross-Border Wedges [Back](#)

τ_{ij}^L — Relationship

- ▶ $\downarrow \mathcal{A}_{bf}^L$
- ▶ Lower match probability
- ▶ PPML on lending relationships: huge

τ_{ij}^R — Pricing

- ▶ $\uparrow \mathcal{A}_{bf}^R$
- ▶ Higher rate conditional on match
- ▶ OLS on loan rates: negligible

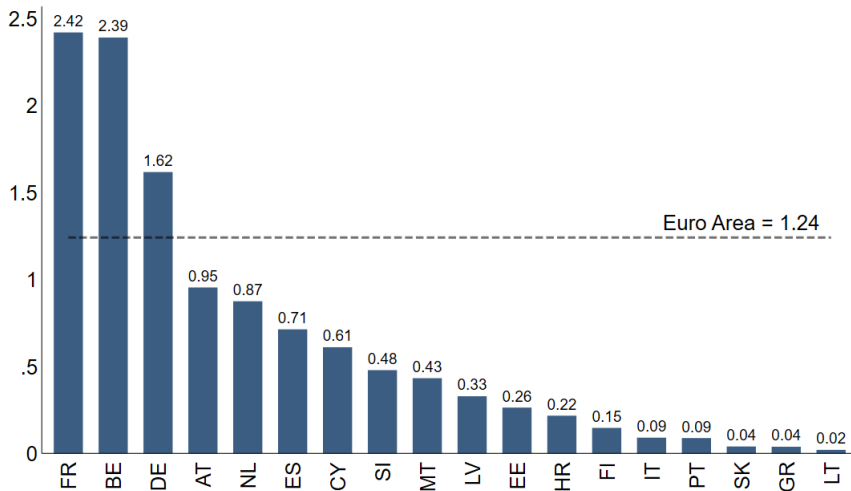
τ_{oi}^E — Entry

- ▶ $\uparrow$ entry cost
- ▶ Fewer foreign branches/subs
- ▶ LPM on group presence: huge

Parameter	Description	Value	Source
κ, λ	Production elasticities	(0.22, 0.64)	Ottonello & Winberry (2020)
δ	Depreciation rate	0.05	Conventional
σ_L	Substitution across lenders	1	Altavilla et al. (2022)
σ_D	Substitution across depositors	1	Tanner (2021)
ψ_3	Frisch elasticity	2.5	Keane & Rogerson (2012)
α	Loss given default	0.04	ECB (2024)

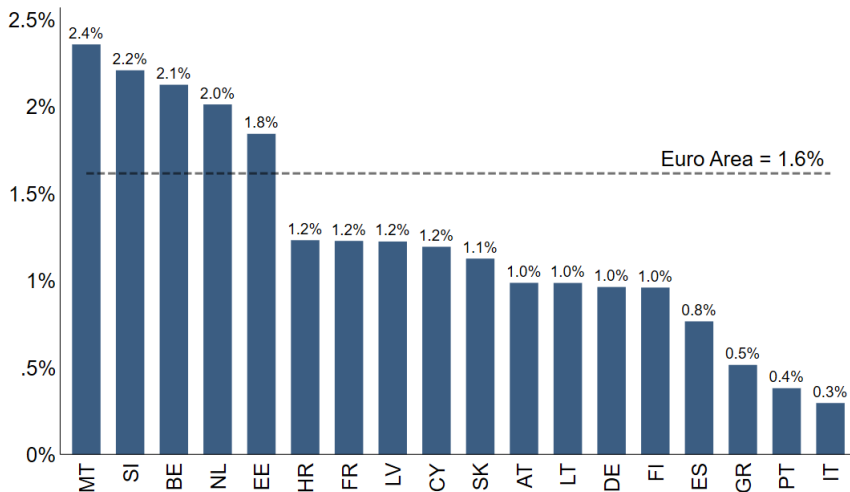
1. Firm productivity and default risk (z_f, G_f)
 - ▶ invert using firms' output and reported proba of default
2. Dimension-reduction: partition firms into deciles of z_f and G_f in each country i
 - ▶ Obtain 20×100 representative firms
3. Bilateral loan-demand and interest rate shifters ($\mathcal{A}_{bf}^L, \mathcal{A}_{bf}^R$)
 - ▶ given u_{bf} estimated from capital-output ratio, match loan shares and interest rates
4. Cross-border wedges (τ^L, τ^R): from microeconomic wedge accounting
5. Cross-border entry wedges (τ^E) and elasticity to bank profits (σ^E)

Integration Raises Foreign Credit Share

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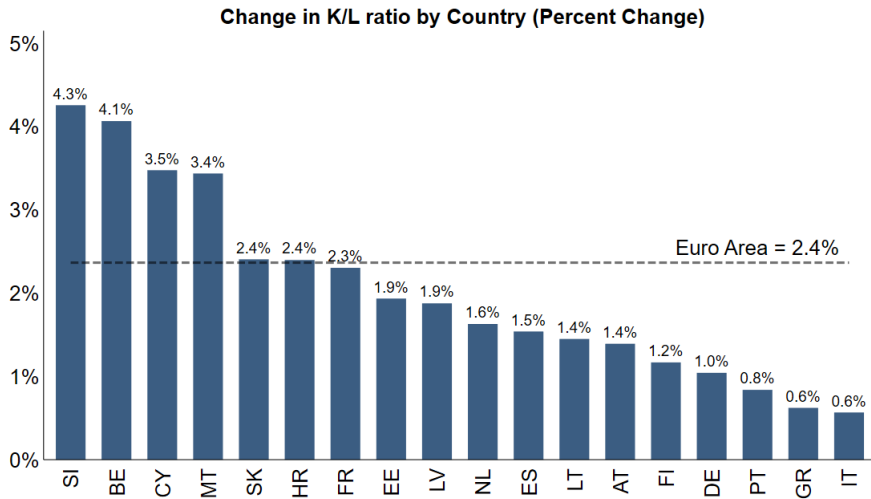
Exercise: 10% reduction in cross-border wedges (τ^L), bank network fixed

Heterogeneous Gains from Integration

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Exercise: 10% reduction in cross-border wedges (τ^L), bank network fixed

Integration Leads to Capital Deepening...

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Exercise: 10% reduction in cross-border wedges (τ^L), bank network fixed

But Limited Improvements in Allocative Efficiency

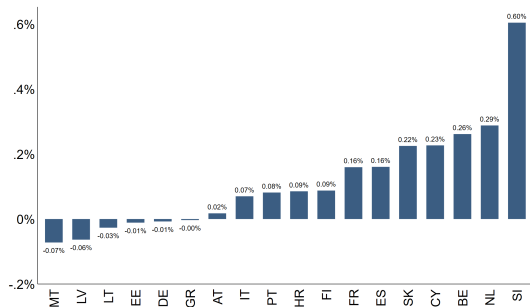
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Figure: $\Delta \ln(MPK)$ by country

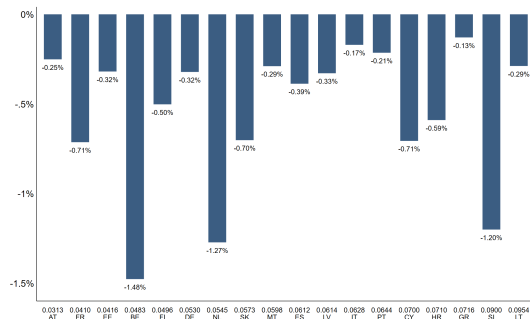


Figure: $\Delta V(\ln(MPK))$ by country